



GRACE COLLEGE OF ENGINEERING
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MULLAKKADU, THOOTHUKUDI - 628 005

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA
SCIENCE

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Unit- I Intelligent Agents

Prepared By,

Mrs.R.Janani, AP/AI&DS

Unit- I Intelligent Agents

PART A

1 Define AI. Or What is AI ?

AI is branch of computer science that deals with the automation of intelligent behaviour. AI gives basis for developing human like programs which can be useful to solve real life problems and thereby become useful to mankind.

2.What is meant by robotic agent ?

A machine that looks like a human being and performs various complex acts of human being. It can do the task efficiently and repeatedly without fault. It works on The basis of a program feeded to it, it can have previously stored knowledge and it can gather more knowledge from environment through its sensors. It acts with the help of actuators.

3. Q1 Define an agent.

An agent is anything (a program, a machine assembly) that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

4. Define rational agent.

A rational agent is agent which always works as per the expected performance. It is a agent who always acts perfectly as per the expectation. It tries to maximize expected behavioural performance.

5. Give the general model of learning agent.

Learning agent model have 4 component-

- 1) Learning Element - Which is responsible for making improvements.
- 2) Performance Elements - Which is responsible for selecting external actions.
- 3) Critic - It tells how agent is doing. (that is it evaluates output).
- 4) Problem Generator It responsible for suggesting actions.

6.What is role of agent program ?

Agent program is important and central part of agent system. It drives the agent, which means that it analyzes data and provides probable actions agent could take.

An agent program is internally implemented as agent function.

An agent program takes input as the current percept from the sensor and return an action to the effectors (Actuator).

7. List down the characteristics of intelligent agent

- 1) The IA must learn and improve through interaction with the environment
- 2) The IA must adapt online and in the real time situation
- 3) The IA must accommodate new problem solving roles incrementally

8 What are the functionalities of the agent function?

Agent function is a mathematical function which maps each and every possible percept sequence to a possible action.

Agent function can be represented in the tabular form

9. Define basic agent program.

The basic agent program is a concrete implementation of the agent function which runs on the agent architecture. Agent program puts forward the lengths of percept sequence and considers only required percept sequences. Agent program implements the functions of percept sequence and action which are external characteristics of the agent.

Agent program takes input as the current percept from the sensor and returns an action to the effectors (Actuators).

10. What are the four components to define a problem? Define them.

The four components to define a problem are,

1. Initial state - It is the state in which agent starts in.
2. A description of possible actions - It is the description of possible actions which are available to the agent.
3. The goal test - It is the test that determines whether a given state is goal (final) state.
4. A path cost function - It is the function that assigns a numeric cost (value) to each path. The problem-solving agent is expected to choose a cost-function that reflects its own performance measure.

11. How will you measure the problem solving performance?

Problem solving performance is measured with 4 factors.

- 1) Completeness Does the algorithm (solving procedure) surely find solution if really the solution exists
- 2) Optimality - If multiple solutions exist then does the algorithm return optimal among them.
- 3) Time requirement.
- 4) Space requirement.

12. What is application of BFS.

It is simple search strategy, which is complete i.e, it surely gives solution if solution exists. If the depth of search tree is small then BFS is the best choice. It useful in tree as well as in graph search.

13 State on which basis search algorithms are chosen.

Search algorithms are chosen depending on two components.

1) How is the state space - That is, state space is tree structured of graph Critical factor for state space is what is branching factor and depth level that tree or graph.

2) What is the performance of the search strategy? A complete, optimal search strategy with better time and space requirement is critical factor performance of search strategy.

14. Evaluate performance of problem solving method based on depth-first-search algorithm.

DFS algorithm performance measurement is done with four ways -

1) Completeness It is complete (guarantees solution).

2) Optimality It is not optimal.

3) Time complexity It's time complexity is $O(b)$.

4) Space complexity It's space complexity is $O(b \cdot d + 1)$.

15. What are the four components to define a problem? Define them.

The four components to define a problem are,

1. Initial state It is the state in which agent starts in.

2. A description of possible actions It is the description of possible actions which are available to the agent.

3. The goal test - It is the test that determines whether a given state in goal(final)

4. A path cost for problems so the function that assigns a numeric cost (value) each path. The problem-solving agent is expected to choose a cost that reflects its own performance measure.

16. Define the bi-directed search.

As the techniques search is bidirectional that is two directional search this searching technique starts the forward search which starts from initial state and the other the searches meet in the middle starts from goal state. The two search state and when both middle.

18. Why problem formulation must follow goal formulation ?

Goal based agent nets the one which solves the problem. Therefore while formulating problem one need to only consider what is the goal to be achieved so that problem formulation is done accordingly. Hence problem formulation must follow goal formulation.

19. Mention how the search strategies are evaluated ?

Search strategies are evaluated on following four criterion

1. completeness: does the search strategy always find a solution, if one exists?
2. time complexity: how much time the search strategy takes to compute.
3. space complexity: how much memory consumption search strategy has ?
4. optimality does the search strategy find the highest-quality solution?

20. Define admissible and consistent heuristics.

Admissible heuristic: A heuristic is admissible if the estimated cost is never more than the actual cost from the current node to the goal node.

Consistent heuristic: A heuristic is consistent if the cost from the current node to a successor node, plus the estimated cost from the successor node to the goal is less than or equal to the estimated cost from the current node to the goal.

21. What is the use of online search agent in unknown environment.

Online search agents operate by interleaving computation and action: first it takes an action, and then it observes the environment and computes the next action. Therefore in unknown environment they became useful as the action is taken first then the environment is observed.

Also, online search is a necessary idea for an exploration problem, where the states and actions are unknown to the agent.

22. List some of the uninformed search techniques.

The uninformed search strategies are those that do not take into account location of the goal. That is these algorithms ignore where they are going until they find a goal and report success. The three most widely used uninformed search strategies are, 1. Depth-First Search - It expands the deepest unexpanded node.

2. Breadth-First Search - It expands the shallowest unexpanded node.

3. Lowest-Cost-First Search (Uniform Cost Search) - It expand the lowest cost node

23. Characterize the environment of the agent playing soccer

The environment of an agent playing soccer can be characterized as dynamic, competitive, and interactive. The agent must adapt to rapidly changing conditions, such as moving teammates, opponents, and the ball. It requires real-time decision-making, spatial awareness, and coordination in a physically demanding setting. Additionally, the environment involves both predictable and unpredictable elements, such as the ball's trajectory and opponents' actions.

24. can an agent that senses only partial information about the state always be perfectly rational?

No, an agent that senses only partial information about the state cannot always be perfectly rational. Perfect rationality requires complete knowledge of the environment to make optimal decisions. With partial information, the agent may make suboptimal choices due to uncertainty or lack of awareness of the full state, leading to imperfect decision-making.

25. Define AI and applications of AI.

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and solve problems like humans.

Applications of AI include:

1. **Healthcare:** AI helps in diagnosing diseases, personalizing treatments, and aiding drug discovery.
2. **Transportation:** AI powers self-driving cars and optimizes traffic management.
3. **Finance:** AI is used in fraud detection, algorithmic trading, and credit scoring.
4. **Retail:** AI personalizes shopping experiences and manages inventory efficiently.

26. List the steps involved in simple problem solving technique.

The steps involved in a simple problem-solving technique in **Artificial Intelligence** are:

1. **Problem Definition:** Clearly define the problem to be solved, including the initial state, the goal, and the possible actions or solutions.
2. **Solution Search:** Explore potential solutions by searching through the problem space, using methods like search algorithms (e.g., breadth-first search, depth-first search).

27. Formulate PEAS for an automated taxi drive

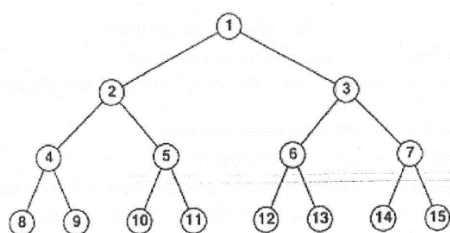
The PEAS (Performance measure, Environment, Actuators, and Sensors) framework helps to define the key components of an intelligent agent. For an automated taxi driver, the PEAS can be formulated as follows:

- **Performance Measure:**
 - Safety of passengers and pedestrians.
 - Timely arrival at destinations.
 - Fuel efficiency or battery usage.
 - Passenger satisfaction (comfort, smoothness of the ride).
 - Cost-effectiveness (optimal route choice, minimal travel time).
- **Environment:**
 - Roads and traffic conditions.
 - Other vehicles, pedestrians, and cyclists.
 - Traffic signals and road signs.
 - Weather conditions (rain, fog, snow, etc.).
 - Pick-up and drop-off locations.
- **Actuators:**
 - Steering mechanism (for steering the vehicle).

- Throttle and brakes (for controlling speed).
- Indicators (for signaling turns or lane changes).
- Horn (for warning pedestrians or other vehicles).
- **Sensors:**
 - Cameras (for detecting objects, pedestrians, and traffic signs).
 - LIDAR (for mapping the surroundings and detecting obstacles).
 - GPS (for location and navigation).
 - Radar (for detecting the speed of other vehicles).
 - Accelerometers and gyroscopes (for monitoring the vehicle's motion and stability).

PART-B

1. Explain the goal and model based reflex agents with example.
2. Explain in detail about the structure of different intelligent agents.
3. What is an agent? Explain the basic kinds of agent program.
4. Explain the properties of Environment.
5. Explain the components of problem definition with example.
6. Discuss uninformed search methods with example.
7. What is uninformed or blind search with examples.
8. Briefly explain the search strategies in uninformed search.
9. Define an agent. Explain the four basic agents that embody the principles underlying intelligent systems with examples.
10. Consider a state space given below



- i) If the goal is numbered '11', list the order in which the states will be visited using (1) Breadth first search
- (2) Depth first search (3) Depth limited search with a limit 2 (depth of the root is 0)
- (4) Iterative deepening search

11. If iterative deepening search is used to traverse the state space, how many times will the state space tree be constructed to reach the goal.

iii) If bidirectional search is used to reach the goal state, what would be the branching factor in the forward and backward direction?

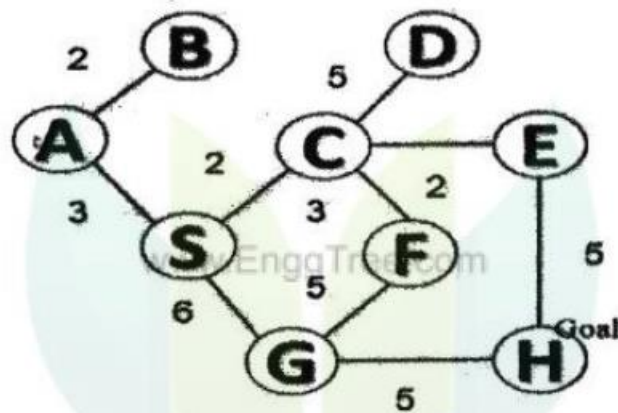
12.What are informed search techniques? Explain any one.

13.List the basic kinds of intelligent agents and explain any two agents with neat diagram.

14.

Consider the water jug problem: You are given two jugs, a 4-gallon one and 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallon of water from the 4-gallon jug? Explain the state space representation and apply an optimal sequence of actions to solve it.

15.



Perform BFS, DFS, Uniform cost search strategies on the following graph and also formulate the algorithm for any 2 strategies.